

Cauchy: $\lim_{x \rightarrow x_0} f(x) = L. \quad \forall \varepsilon > 0 \Rightarrow \boxed{\exists \delta > 0 \quad 0 < |x - x_0| < \delta}$

$\Rightarrow \quad |f(x) - L| < \varepsilon.$

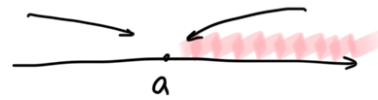
- 1) $\forall \varepsilon > 0$
- 2) $|f(x) - L| \leftarrow \exists \delta > 0.$
- 3) $\{ \varepsilon \} \rightarrow \left[\frac{\varepsilon}{n} \right] \quad \delta = \left[\frac{\varepsilon}{n} \right]$

Heine: $\lim_{n \rightarrow \infty} f(x_n) = L$; $f(x)$ $\rightarrow \{x_n\} \rightarrow \boxed{\lim_{n \rightarrow \infty} x_n = x_0}$

$\rightarrow \bullet \lim_{n \rightarrow \infty} f(x_n) = L.$

$$\lim_{x \rightarrow 0} \frac{1}{x} = \infty \quad (\text{DNE})$$

$$\lim_{x \rightarrow \infty} \frac{1}{x} = 0$$



Problem 1a

a) $\boxed{\lim_{x \rightarrow a^+} f(x) = -\infty}$

right-hand limit: $x \rightarrow a^+$ we consider only points to the right of a : $\underline{a} < x < \underline{a + \delta}$

Def $\forall \varepsilon > 0$; $\forall M > 0 \quad \exists \delta > 0 \quad \forall x \in (a; a + \delta):$
 $f(x) \leq -M$

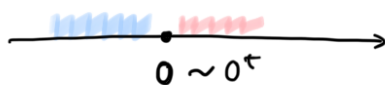
negation: $\nexists \leftrightarrow \exists$
 $> \sim <$

Negation: $\lim_{x \rightarrow a^+} f(x) \neq -\infty \Leftrightarrow \forall \varepsilon > 0; \exists M > 0, \forall \delta > 0$
 $\exists x \in (a; a + \delta) : f(x) \geq -M.$

Problem 2.

a), b) $\lim_{x \rightarrow 0^+} \frac{x}{|x|} = ?$ $\lim_{x \rightarrow 0^-} \frac{x}{|x|} = ?$

$$|x| = \begin{cases} x, & x > 0 \\ -x, & x < 0 \end{cases}$$



$$0, \quad 0^+ = 0.000001$$

$$0, \quad 0^- = -0.000000...1$$

$$\lim_{x \rightarrow 0^+} \frac{x}{|x|} = \lim_{x \rightarrow 0^+} 1 = 1. \quad \checkmark$$

$$\lim_{x \rightarrow 0^-} -1 = -1. \quad \checkmark$$

$$c), d) \quad \lim_{x \rightarrow 2^+} \frac{x^2 - x - 1}{x^2 - 5x + 6} = \frac{4 - 2 - 1}{4 - 10 + 6} = \left\{ \frac{1}{0} \right\} =$$

$$= \lim_{x \rightarrow 2^+} \left[\frac{x^2 - x - 1}{\underbrace{(x-2)}_{\substack{2^+-2 \\ 0^+}} \underbrace{(x-3)}_{(-1)}} \right] = \frac{1}{\underbrace{0^+ \cdot (-1)}_{(-0)}} = -\infty$$

$$\lim_{x \rightarrow 2^-} \frac{x^2 - x - 1}{x^2 - 5x + 6} = \left\{ \frac{1}{0} \right\} = \lim_{x \rightarrow 2^-} \frac{x^2 - x - 1}{\underbrace{(x-2)}_{\substack{2^--2 \\ 0^-}} \underbrace{(x-3)}_{(-1)}} = \frac{1}{\underbrace{0^- \cdot (-1)}_{+0}} = +\infty$$

Problem 3

$$f(x) = \begin{cases} |1-x|, & x < 2 \\ \sqrt{x+2}, & x \geq 2. \end{cases}$$

$$\lim_{x \rightarrow 2^-} = ?$$

$$\lim_{x \rightarrow 2^+} = ?$$

$$x \rightarrow 2^- : f(x) = |1-x|$$

$$x \rightarrow 2^+ : f(x) = \sqrt{x+2}$$

$$\lim_{x \rightarrow 2^-} |1-x| = |1-2| = 1$$

$$\lim_{x \rightarrow 2^+} \sqrt{x+2} = \sqrt{2+2} = 2$$

$$\lim_{x \rightarrow 2^-} \neq \lim_{x \rightarrow 2^+}$$

$\Rightarrow$ function is not cont $\Rightarrow$ the lim, where $x \rightarrow 2$ DNE.

Problem 4.

$$\lim_{x \rightarrow \infty} \left(\frac{x^2 + 1}{x + 1} - ax - b \right) = 0, \quad a, b = ?$$

$$\lim_{x \rightarrow \infty} f(x) = A$$

$$\lim_{x \rightarrow \infty} h(x) = B$$

$$\lim_{x \rightarrow \infty} (f(x) - h(x)) = A - B$$

$$\text{as } x \rightarrow \infty : \frac{c}{x + \dots} = 0$$

$$\frac{x^2 + 1}{x + 1} = \frac{x^2 + 1}{x^2 + x} \cdot \frac{x + 1}{x - 1} = x - 1 + \frac{2}{x + 1}$$

$$\lim_{x \rightarrow \infty} \left(\boxed{x-1} + \frac{\frac{-1-x}{2}}{\boxed{x+1}} \boxed{-ax-b} \right) = 0$$

$$\lim_{x \rightarrow \infty} \left(x \underbrace{(1-a)}_0 - (1+b) \right) = 0^*$$

$$\Rightarrow 1-a=0 \Rightarrow a=1$$

$$-(1+b)=0 \Rightarrow b=-1$$

Ans: $a=1$, $b=-1$.

Problem 5.

$$\lim_{x \rightarrow 1} \frac{x^m - 1}{x^n - 1} = \frac{1^m - 1}{1^n - 1} = \left\{ \frac{0}{0} \right\} =$$

$$\boxed{x^k - 1} = (x-1)(x^{k-1} + x^{k-2} + \dots + \underbrace{x^{k-k}}_1)$$

$$\underline{\underline{\sqrt{x^3 - 1}}} = (x-1)(x^2 + x^1 + 1) = \underbrace{x^3 + x^2 + x - x^2 - x - 1}_m$$

$$= \lim_{x \rightarrow 1} \frac{(x-1)(x^{m-1} + x^{m-2} + \dots + 1)}{(x-1)(x^{n-1} + x^{n-2} + \dots + 1)} = \frac{m}{n}$$

Problem 9.

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

→ 1st remarkable limit.

$$\frac{\sin x}{dx}$$

$$\sin^2 x + \cos^2 x = 1$$

$$\sin x = 2 \cos \frac{x}{2} \sin \frac{x}{2}$$

$$1 - \cos x = 2 \sin^2 \frac{x}{2}$$

$$\sin u - \sin v = 2 \sin \frac{u-v}{2} \cos \frac{u+v}{2}$$

$$a) \lim_{x \rightarrow 0} \frac{\sin 3x}{x} = \lim_{x \rightarrow 0} \frac{\sin 3x}{3 \cdot x} = 3.$$

$$b) \lim_{x \rightarrow 0} \frac{\sin 7x}{\sin 8x} = \lim_{x \rightarrow 0} \frac{7x}{\frac{\sin 8x}{8x} \cdot 8x} = \frac{7}{8}$$

$$c) \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{x^2} =$$

$$\left(\frac{\sin \left(\frac{x}{2} \right)}{\frac{x}{2}} \right)^2 = \boxed{\frac{4 \sin^2 \left(\frac{x}{2} \right)}{x^2}}$$

$$= 2 \cdot \lim_{x \rightarrow 0} \frac{\sin^2 \left(\frac{x}{2} \right)}{x^2} = 2 \cdot \lim_{x \rightarrow 0} \frac{4 \sin^2 \left(\frac{x}{2} \right)}{4x^2} =$$

$$= 2 \cdot \frac{1}{4} = \frac{1}{2}$$

$$d) \lim_{x \rightarrow a} \frac{\sin x - \sin a}{x - a} = \left\{ \frac{0}{0} \right\};$$

$$\sin x - \sin a = 2 \sin \frac{x-a}{2} \cos \frac{x+a}{2}, \quad x=a=0.$$

$$\lim_{x \rightarrow a} \frac{\frac{1}{2} \cdot 2 \sin \left(\frac{x-a}{2} \right) \cos \left(\frac{x+a}{2} \right)}{\frac{x-a}{2}} = \cos \frac{a+a}{2} = \boxed{\cos a}$$

$$f) \lim_{x \rightarrow \pi} \frac{\sin x}{\pi^2 - x^2} = \left\{ \frac{0}{0} \right\} \rightarrow \begin{cases} \text{Subst:} \\ y = \pi - x, & y \rightarrow 0 \\ x = \pi - y \end{cases}$$

$$= \lim_{y \rightarrow 0} \frac{\sin(\pi - y)}{(\pi - \pi + y)(\pi + \pi - y)} = \lim_{y \rightarrow 0} \frac{\sin(\pi - y)}{y(2\pi - y)} =$$

$$= \lim_{y \rightarrow 0} \frac{\sin y}{y(2\pi - y)} = \lim_{y \rightarrow 0} \frac{1}{2\pi - y} = \frac{1}{2\pi}$$